SIMATS ENGINEERING

ASSESSMENT TOOL 2: Schema Analysis and Viva Voce

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 35%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 20**
Question Count: 2

Code of Conduct

I **UDHAYAKUMAR U (192424332)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: U. Udhaya Kum

Database Design and Connectivity Rubric

Requirement Analysis (4 Marks)	ER Diagram Design (4 Marks)	Table Design & Normalization (4 Marks)	Database Connectivity using Python (4 Marks)	CRUD Operations (4 Marks)	Total (20 Marks)

Total Marks Awarded:

Questions:

Question 1: Student Course Registration Database (10

Marks) A university maintains the following database

schema:

- Student (Student_ID, Student_Name, Department, Year)
- Course (Course_ID, Course_Name, Credits)
- Enrollment (Enrollment_ID, Student_ID, Course_ID, Semester)

Aim

To develop a Python program using SQLite that creates a Student Course Registration database, inserts sample data, performs an SQL JOIN operation, and displays the names of students along with the courses they are enrolled in during the Odd 2026 semester.

Algorithm

- Import the sqlite3 module.
- Connect to the SQLite database.
- Create the tables:
- Student
- Course
- Enrollment
- Insert sample records into all tables.
- Execute an SQL JOIN query to combine Student, Course, and Enrollment tables.
- Filter the records where Semester = 'Odd 2026'.
- Display Student Name and Course Name.
- Close the database connection.

Python Code (SQLite)

```
import sqlite3
# Connect to SQLite databaseconn = sqlite3.connect("student.db")cur = conn.cursor()
# Create tablescur.execute("""
CREATE TABLE IF NOT EXISTS Student(
Student_ID INTEGER PRIMARY KEY,
Student_Name TEXT,
Department TEXT,
Year INTEGER)
""")
cur.execute("""
CREATE TABLE IF NOT EXISTS Course(
Course_ID INTEGER PRIMARY KEY,
Course_Name TEXT,
Credits INTEGER)
""")

cur.execute("""
CREATE TABLE IF NOT EXISTS Enrollment(
Enrollment_ID INTEGER PRIMARY KEY,
Student_ID INTEGER,
Course_ID INTEGER,
Semester TEXT,
FOREIGN KEY(Student_ID) REFERENCES Student(Student_ID),
FOREIGN KEY(Course_ID) REFERENCES Course(Course_ID))
""")

# Insert sample datacur.execute("INSERT INTO Student VALUES
(1,'Arun','CSE',2)")cur.execute("INSERT INTO Student VALUES (2,'Priya','IT',3)")
cur.execute("INSERT INTO Course VALUES (101,'Python
Programming',4)")cur.execute("INSERT INTO Course VALUES (102,'Database Systems',3)")
cur.execute("INSERT INTO Enrollment VALUES (1,1,101,'Odd 2026')")cur.execute("INSERT
INTO Enrollment VALUES (2,2,102,'Odd 2026')")
conn.commit()

# JOIN Query-query = """
SELECT Student.Student_Name, Course.Course_Name
FROM Enrollment
JOIN Student
ON Enrollment.Student_ID = Student.Student_ID
JOIN Course
ON Enrollment.Course_ID = Course.Course_ID
WHERE Enrollment.Semester='Odd 2026'
"""
```

```
print("Students enrolled in Odd 2026 Semester\n")
for row in cur.execute(query):
    print("Student:", row[0], " Course:", row[1])
conn.close()
```

OUTPUT

```
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

=== RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/sqlite.py ==
Students enrolled in Odd 2026 Semester

Student: Arun  Course: Python Programming
Student: Priya  Course: Database Systems
```

Question 2: Hospital Management Database (10

Marks) A hospital maintains the following

database schema:

- Doctor (Doctor_ID, Doctor_Name, Specialization)
- Patient (Patient_ID, Patient_Name, Age)
- Appointment (Appointment_ID, Patient_ID, Doctor_ID, Appointment_Date)

Analyze the schema and write a Python program to connect to the database and display the patient names, doctor names, and appointment dates for all appointments scheduled after 1 January 2026.

Aim

To develop a Python program using **SQLite** that creates a Hospital Management database, inserts sample data, and displays patient names, doctor names, and appointment dates for appointments scheduled **after 1 January 2026**.

Algorithm

- Import the sqlite3 module.
- Connect to the SQLite database.
- Create the tables:

    1. Doctor
    2. Patient
    3. Appointment

- Insert sample data.
- Execute an SQL JOIN query on the three tables.
- Filter appointment dates greater than '2026-01-01'.
- Display Patient Name, Doctor Name, and Appointment Date.
- Close the database connection.

Python Code (SQLite)

```
import sqlite3
# Connect to SQLite databaseconn = sqlite3.connect("hospital.db")cur = conn.cursor()
# Create tablescur.execute("""
CREATE TABLE IF NOT EXISTS Doctor(
Doctor_ID INTEGER PRIMARY KEY,
Doctor_Name TEXT,
Specialization TEXT)
""")

cur.execute("""
CREATE TABLE IF NOT EXISTS Patient(
Patient_ID INTEGER PRIMARY KEY,
Patient_Name TEXT,
Age INTEGER)
""")

cur.execute("""
CREATE TABLE IF NOT EXISTS Appointment(
Appointment_ID INTEGER PRIMARY KEY,
Patient_ID INTEGER,
Doctor_ID INTEGER,
Appointment_Date TEXT,
FOREIGN KEY(Patient_ID) REFERENCES Patient(Patient_ID),
FOREIGN KEY(Doctor_ID) REFERENCES Doctor(Doctor_ID))
""")

# Insert sample datacur.execute("INSERT INTO Doctor VALUES(1,'Dr.
Kumar','Cardiology')")cur.execute("INSERT INTO Doctor VALUES(2,'Dr.
Meena','Orthopedics')")
cur.execute("INSERT INTO Patient VALUES(1,'Rahul',25)")cur.execute("INSERT INTO
Patient VALUES(2,'Anitha',30)")
cur.execute("INSERT INTO Appointment VALUES(1,1,1,'2026-02-
15')")cur.execute("INSERT INTO Appointment VALUES(2,2,2,'2026-03-20')")
conn.commit()

# JOIN Queryquery = """
SELECT Patient.Patient_Name,
Doctor.Doctor_Name,
Appointment.Appointment_Date
FROM Appointment
JOIN Patient
ON Appointment.Patient_ID = Patient.Patient_ID
JOIN Doctor
```

```
ON Appointment.Doctor_ID = Doctor.Doctor_ID
WHERE Appointment.Appointment_Date > '2026-01-01'
"""
print("Appointments After 1 January 2026\n")
for row in cur.execute(query):
    print("Patient:", row[0],
          " Doctor:", row[1],
          " Date:", row[2])
conn.close()
```

OUTPUT

```
>>>
=== RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/sqlite.py ==
Appointments After 1 January 2026

Patient: Rahul  Doctor: Dr. Kumar  Date: 2026-02-15
Patient: Anitha  Doctor: Dr. Meena  Date: 2026-03-20
>>>
```

Signature of the Course Faculty